

TREDENCE

CAPSTONE



AI Sales Deal Prediction & Recommendation Agent

#BeyondPossible

Predicting Win Probability and Recommending the Next Best Action for Sales Deals

Agenda

AI Sales Deal Prediction & Recommendation Framework

01 BUSINESS CONTEXT

- Problem & Objective
- Project Scope

02 DATA & MODELING

- Data Preparation
- Model Development

03 PREDICTIONS

- Win Probability & Expected Revenue
- Deal Recommendations

04 RESULTS & NEXT STEPS

- Model Performance
- Dashboard Demo & Key Insights



Business Problem & Approach

How likely is a given sales opportunity to close as a win, and what should the sales team do about it?

Chasing Losing Deals

Overestimating a deal's win chance wastes sales effort, discounting, and account time on opportunities unlikely to close.

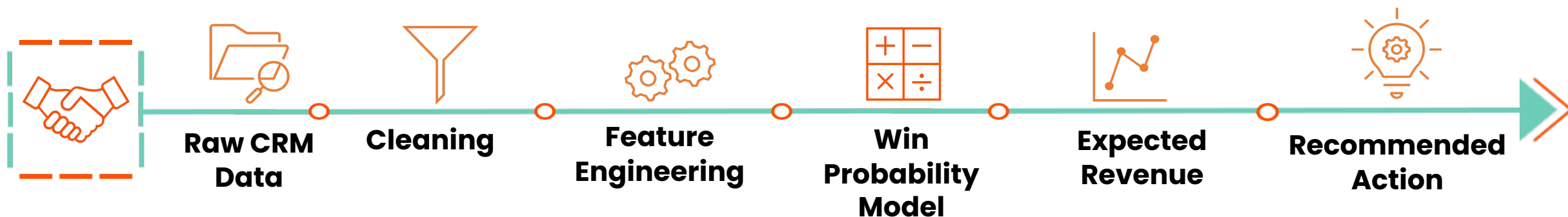
Missing Winnable Deals

Underestimating a deal's win chance can mean it gets deprioritized, under-resourced, or lost to a competitor unnecessarily.

What the Model Delivers

A data-driven win probability score for each deal, its expected revenue, and an actionable recommendation.

AI Sales Deal Prediction Framework



- Model will predict the likelihood a sales opportunity will close as a win
- AI recommends the next best action using historical deal data and sales-process signals



Scope of the Model

IN SCOPE



Unit of Analysis
One Sales Opportunity



Coverage
IBM Watson Sales Win-Loss Public Dataset



Prediction Point
At Current Sales Stage, Before the Deal Closes



Output
Win Probability + Expected Revenue + Recommendation

OUT OF SCOPE



Live CRM Integration
No live CRM connection for real-time scoring



Partner & RFP Data
Partner, practice area, and RFP fields are not in the public dataset



Pipeline Forecasting
Deal value is known at quote time; future pipeline value is not forecasted



Tredence CRM Data
The model uses a public proxy dataset rather than Tredence CRM data

Feature Engineering Formulas

- Nine features feed one LightGBM model that outputs a single headline number, including eight raw signals and one feature we engineered ourselves
- Four more engineered signals are shown on the dashboard for context, but don't feed the model

WIN PROBABILITY

0% – 100%

Likelihood this deal closes as a win

Low / Moderate / Likely to close

Bounced A Lot (dashboard only)

Yes / No

Sales Stage Change Count > 3 – Deal has bounced between stages repeatedly

Is Existing Client (dashboard only)

Yes / No

Revenue From Client Past Two Years > 0 – Client has spent with us before

Is Stalled (dashboard only)

Yes / No

Elapsed Days In Sales Stage > 65 – Deal has sat in its current stage too long

High Risk Deal (dashboard only)

Yes / No

Is Stalled AND Amount > 75th percentile – Closes at only 16.3% vs 22.9% for others

Stage Bounce Rate (MODEL INPUT)

0 – 1

Stage Change Count / (Elapsed Days + 1) – The only engineered feature the model sees

Worked Example

One real deal from our test set, showing from raw facts to the model's prediction



Opportunity Number	Opportunity Amount USD	Elapsed Days In Sales Stage	Sales Stage Change Count	Revenue From Client Past Two Years	Route To Market	Supplies Subgroup	Client Size by Revenue	Competitor Type
6683358	\$300,000	88	2	0	Fields Sales	Batteries & Accessories	1	Unknown

SIGNALS

Stalled	Yes
Bounced	No
Existing Client	No
High Risk	Yes
Stage Bounce Rate	0.0225

MODEL

Predicted Win Probability: 0.8%

EXPECTED REVENUE

Expected Revenue: \$300,000
(Opportunity Amount USD) × 0.8% (Win Probability) = \$2,263

ACTION

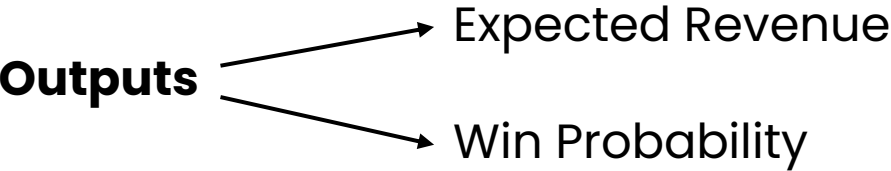
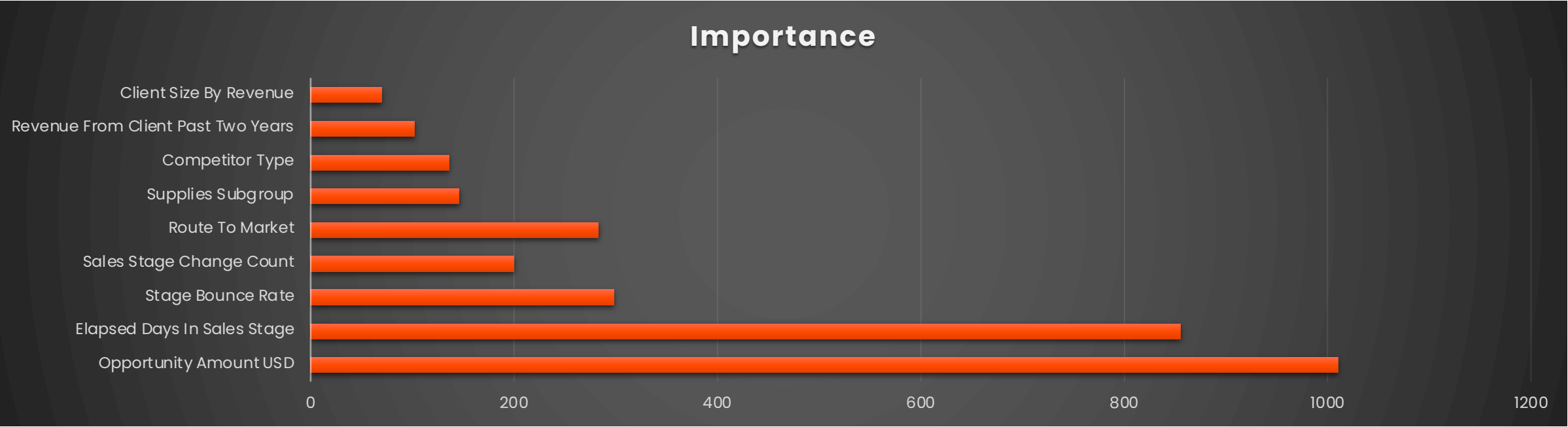
AI Recommendation: Escalate to sales manager within 48 hours

REVEAL

LOST
Model called it right, before we ever knew

Initial Findings

- What the data showed us before we even built the recommendation engine
- Deal amount and time stuck in a stage matters the most, whereas everything else plays a smaller role



Models	AUC	Accuracy
LightGBM	0.903	86.4%
Random Forest	0.872	85.0%
Logistic Regression	0.724	78.7%

Accuracy & Validation

Tested on 15,594 deals the model never saw during training

Accuracy

86.4 %

AUC-ROC

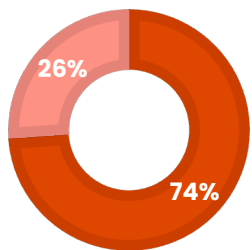
0.903

Precision / Recall

77.9% / 55.5%

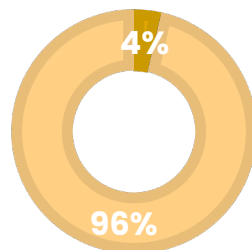
Every deal in the test set falls into one of these 4 outcomes:

**CORRECTLY PREDICTED
VALUE**



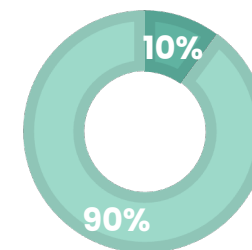
Correctly Predicted Loss

FALSE ALARM



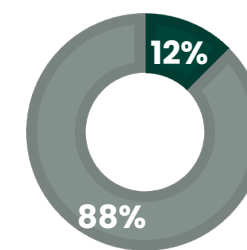
Wrongly predicted Loss

MISSED OPPORTUNITY



Wrongly predicted win

**CORRECTLY PREDICTED
VALUE**



Correctly predicted...

The model was right about 86.4% of the time overall. It occasionally misses a real win (10.1% of deals) and occasionally raises a false alarm on a deal that was going to be lost (3.6% of deals)

Assumptions, Exclusions, & Limitations

Future Behaves like the past and it won't catch sudden market shifts until retrained

Only 1 of 5 new signals helps the model, and Real Tredence details (RFP, team) are not in this data

Trained on public data (not Tredence's own deals) and no automatic connection to a live CRM, so someone must manually add new deals

Business value & Recommended use cases:

Prevents a \$1.18B forecasting error — naive total says \$1.44B, honest number is \$261M.

Saves \$2.5M in wasted sales effort ; stops reps chasing 2,500 deals that are already lost.

Catches problems early enough to still act on them — flags risk while a deal can still be saved, not after it's already lost.

Any deal flagged as stalled or high-risk should trigger a mandatory manager check-in within 48 hours

Leadership should plan quarterly targets around the \$261M expected revenue figure, not the \$1.44B naive total

Next Steps:

Retrain on Tredence's real historical deal data, once available, instead of the public dataset

Connect to a live CRM feed instead of saved files, so it updates automatically as deals move

Expand the rule engine with input from real sales managers on what "urgent" should mean in practice

Live Demo

Opportunity Number

7050675

Deal Amount

\$39,473

Client Size

1

Route To Market

Reseller

Competitor Type

Unknown

Predicted Win Probability

12.1%

PREDICTIVE

Expected Revenue

\$4,781

PREDICTIVE

Deal Status

Stalled

Deal Signals

These 4 flags don't feed the model itself, since they added zero predictive value there. They're kept purely as clear, human-readable business signals.

Stalled

YES

Bounced A Lot

NO

Existing Client

NO

High Risk Deal

NO

Recommended Action

Deal #7050675, worth \$39,473

DEAL STATUS: NEEDS FOLLOW-UP

CATCH-UP SUMMARY

No stage progress in 73 days.

RECOMMENDED ACTION (AI)

Follow up with client

FULL AI ANALYSIS

The numbers unequivocally show that this deal is stalled, with a mere 12% predicted win probability and a prolonged stay of 73 days in the current sales stage. This stagnation is likely due to a lack of recent engagement, as evidenced by the deal's extended duration in its current stage and the fact that it has bounced between stages twice. The clearest recommended action is to follow up with the client within the next 5 days, as mandated by the existing rule-based signal, to reinvigorate the sales process and potentially salvage the deal. If no action is taken, the deal will likely continue to languish, resulting in a further decline in win probability and a potential loss of the expected \$4,781 in revenue. The fact that the client is new and the sales channel is a reseller adds a layer of urgency to the situation, as failure to close the deal may damage the relationship and hinder future sales opportunities. By taking immediate action, we can attempt to rescue this deal and prevent a significant loss of potential revenue.

Send Follow-up Email

Add Follow-Up to Calendar (Aug 16)

AI Sales Deal Agent

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Thank You

#BeyondPossible

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CONTACT DETAILS

Mobile:

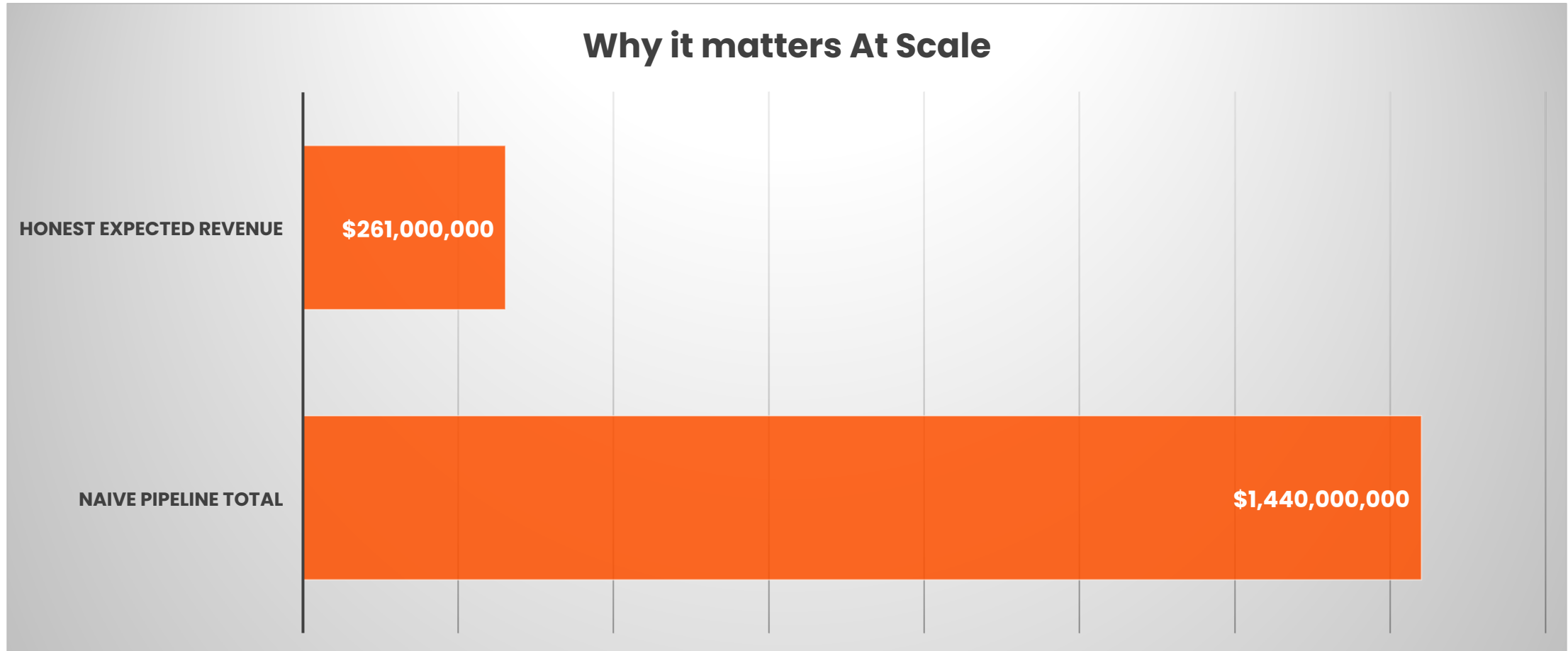
Email:

Recommendation engine: rules + LLM narration

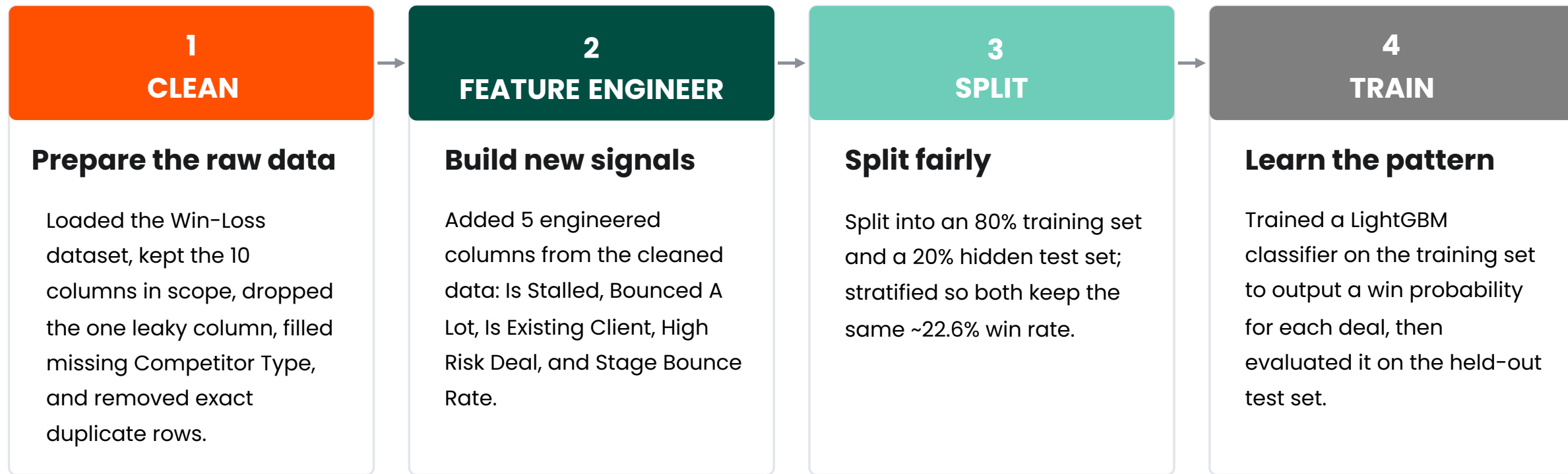
	Rules (The decision maker)	AI / Groq (The narrator)
Job	Decides which of 6 categories a deal falls into	Turns that category into a natural sentence a person would read
Looks at	Is it stalled? Bounced a lot? Is it a big deal? Win probability?	The category the rules already picked + the deal's real numbers
Can it invent anything?	No — just fixed if/then logic	No — told explicitly to only describe real numbers, never make things up
Example output	"High value deal stalled — escalate to manager"	The numbers unequivocally show that this \$300,000 deal has been stagnant for 88 days, with a dismal predicted win probability of 1%, resulting in a meager expected revenue of \$2,263. This deal is in this state due to a clear lack of momentum, having bounced between stages twice without making meaningful progress. The clearest recommended action is to immediately escalate this high-value deal to a manager and schedule an executive call within 48 hours, as dictated by the existing rule-based signal. If no action is taken, this deal will likely continue to languish, ultimately resulting in a lost opportunity and a significant revenue shortfall. Failing to take decisive action will only serve to further diminish the already slim chances of winning this deal. By escalating to a manager and scheduling an executive call, we can inject much-needed momentum into this stalled deal and potentially salvage a significant revenue opportunity.
What if it fails?	Can't fail — it's just math	If the AI is out of token / unavailable, rules (decision maker) is the fallback for it.



The \$1.1B+ gap is exactly how much a naive forecast overstates reality



How the data is prepared



- Everything downstream of Clean is fully reproducible
- The same input data always produces the same features, split, and trained model.

